

PUNTA GORDA, FL, USA

WMO: 722034

Lat: 26.917N

Lon: 81.991W

Elev: 8

StdP: 101.23

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 12812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	6.1	-2.9	3.0	10.7	-0.9	3.5	11.2	10.8	20.8	9.3	20.2	2.5	20	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	34.1	24.7	33.5	24.7	32.9	24.7	26.7	30.8	26.3	30.6	26.0	30.2	3.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.0	25.2	20.4	27.8	24.9	20.0	27.7	82.9	30.5	81.4	30.5	80.1	30.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.0	7.3	DB	-0.1	35.5	1.6	1.0	-1.3	36.2	-2.3	36.8	-3.2	37.3	-4.4	38.0	(4)
			WB	-1.4	27.5	1.9	0.7	-2.8	28.0	-3.9	28.4	-4.9	28.7	-6.3	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	16.6	18.4	20.2	22.7	25.2	27.4	28.1	28.3	27.6	24.9	20.8	18.7	
	DBStd	4.92	4.29	3.91	3.26	2.48	1.83	1.24	1.06	1.05	1.21	2.73	3.31	4.14	
	HDD10.0	8	5	1	0	0	0	0	0	0	0	0	0	2	
	HDD18.3	218	84	43	21	3	0	0	0	0	0	2	15	49	
	CDD10.0	4849	209	237	315	380	472	523	561	567	528	461	325	271	
	CDD18.3	2019	30	47	78	133	214	273	302	309	278	205	90	60	
	CDH23.3	18866	216	368	666	1254	2188	2664	3035	3148	2574	1728	640	384	
	CDH26.7	7247	22	62	141	411	939	1192	1294	1329	1039	620	134	64	
Wind	WSAvg	3.3	3.5	3.5	3.8	3.7	3.5	3.0	2.6	2.6	2.8	3.4	3.4	3.4	
Precipitation	PrecAvg	1308	55	51	63	58	77	235	203	212	194	82	39	37	
	PrecMax	1889	274	105	221	144	196	480	329	355	366	276	131	142	
	PrecMin	896	0	1	5	7	2	117	116	117	115	16	0	6	
	PrecStd	207	55	30	49	37	50	93	63	59	66	60	32	35	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.5	30.0	30.5	32.6	34.2	35.0	34.7	34.9	34.0	33.2	30.8	29.6	
		MCWB	21.2	21.2	20.8	22.3	22.4	24.3	25.1	25.5	24.9	24.6	23.9	22.8	
	2%	DB	27.2	28.5	29.2	31.3	33.0	33.9	33.8	33.9	33.1	32.1	29.3	28.3	
		MCWB	20.4	21.1	20.7	21.8	22.5	24.4	25.2	25.5	25.0	24.3	22.8	22.0	
	5%	DB	26.0	27.3	28.2	30.3	32.3	33.0	33.0	33.1	32.5	31.1	28.2	27.2	
		MCWB	19.8	20.4	20.7	21.4	22.5	24.4	25.2	25.5	25.0	23.9	21.8	21.4	
	10%	DB	24.1	26.0	27.3	29.1	31.2	32.2	32.3	32.4	31.8	30.1	27.2	25.8	
		MCWB	18.9	19.8	20.2	21.0	22.5	24.5	25.3	25.5	25.0	23.3	20.8	20.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	23.2	24.3	24.7	25.2	26.8	26.9	27.3	27.0	26.1	25.2	24.3
			MCDB	26.1	27.1	27.7	29.4	30.6	31.2	31.1	30.9	30.6	29.7	28.5	27.7
2%		WB	21.7	22.4	23.3	23.9	24.7	26.0	26.4	26.8	26.4	25.6	24.1	23.3	
		MCDB	25.1	26.4	26.5	28.7	29.8	30.8	30.9	30.7	30.4	29.5	27.7	26.4	
5%		WB	20.9	21.6	22.4	23.0	24.2	25.6	26.0	26.4	25.9	25.1	23.2	22.4	
		MCDB	24.3	25.5	25.8	27.6	29.1	30.4	30.5	30.5	29.9	29.1	26.4	25.5	
10%		WB	20.0	20.8	21.4	22.4	23.8	25.2	25.7	26.0	25.6	24.6	22.2	21.5	
		MCDB	23.2	24.5	25.1	26.9	28.6	30.0	30.2	30.1	29.7	28.3	25.4	24.5	
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.7	12.8	12.8	12.4	9.8	8.8	8.7	8.7	10.3	11.7	11.8	
		MCDBR	12.9	12.9	13.3	13.7	13.5	11.3	9.7	9.5	9.5	10.3	11.4	11.7	
		MCWBR	6.7	6.3	6.0	5.6	4.8	3.8	3.1	3.1	3.1	3.7	5.2	5.8	
	5% WB	MCDBR	11.1	11.1	10.5	11.5	11.1	9.3	8.7	8.6	8.5	8.8	9.8	10.0	
		MCWBR	6.7	6.0	5.8	5.6	4.4	3.6	3.2	3.1	3.1	3.4	5.0	5.7	
Clear-Sky Solar Irradiance	taub	0.347	0.361	0.375	0.391	0.419	0.443	0.461	0.448	0.422	0.397	0.359	0.350		
	taud	2.512	2.481	2.431	2.359	2.336	2.347	2.291	2.355	2.404	2.443	2.493	2.528		
	Ebn at Noon	903	915	919	907	876	849	833	845	861	865	880	883		
	Edh at Noon	93	103	114	126	129	127	133	125	116	105	94	88		
All-Sky Solar Radiation	RadAvg	3.68	4.48	5.50	6.44	6.68	6.01	5.87	5.66	5.12	4.75	3.97	3.39		
	RadStd	0.27	0.27	0.37	0.27	0.43	0.52	0.38	0.31	0.31	0.28	0.20	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.63	+1.45	+1.86	N/A	N/A	N/A	-4	N/A	+248	+212	(46)	
(47)	Regional (31 neighbors)	+0.42	+0.82	N/A	N/A	N/A	N/A	N/A	-35	+158	+116	(47)	

Nomenclature: See separate page